Experiment 5: Model Training, Hyperparameter Tuning & Cross-Validation

CSE4001: Applications of Machine Learning in Industry — Health-Project 3 (Breast Cancer)

Course: CSE4001 Applications of ML	**Dataset:** WDBC (569 FNA Samples, 30 Features)
Focus: Model Training & Tuning	**Partitions:** Train (80%, N=455), Val (10%, N=57), Test (10%, N=57)
Algorithms: Logistic Reg., SVM (RBF), Random Forest	**Status:** Completed & Verified

1. Aim

To implement, train, hyperparameter-tune, and validate candidate machine learning classification algorithms (**Logistic Regression**, **Support Vector Machine**, and **Random Forest**) on the standardized Breast Cancer Wisconsin dataset using Stratified 5-Fold Cross-Validation, compare model performance against baseline heuristics, and serialize the optimal model checkpoints for downstream clinical evaluation.

2. Objectives

- **Establish Baseline Heuristics:** Implement a Zero-Rule Dummy classifier (majority class heuristic) and an untuned Logistic Regression baseline.
- **Train Candidate Classifiers:** Implement Regularized Logistic Regression (L2), Support Vector Classifier (RBF kernel), and Random Forest (ensemble bagging).
- **Hyperparameter Optimization:** Execute Stratified GridSearchCV (k=5) across regularizers (C), kernel coefficients (γ), and tree depths.
- **Evaluate Multi-Metric Generalization:** Track balanced accuracy, malignant recall (sensitivity), specificity, precision, F1-score, and ROC-AUC.
- **Error & Confusion Matrix Analysis:** Evaluate models on the independent validation holdout set (N=57) to minimize False Negatives (missed cancer).
- **Feature Attribution & Interpretability:** Compare Random Forest Mean Decrease in Impurity (Gini) with Logistic Regression standardized weights.
- **Model Checkpoint Serialization:** Save trained model artifacts (.pkl) and training metadata JSON in the models directory.

3. Mathematical Formulations of Candidate Algorithms

Model Architecture	Mathematical Formulation & Objective Function	Key Hyperparameters
Logistic Regression (L2)	$P(y=1\|x) = \sigma(w^T x + b) = 1 / (1 + e^{-(w^T x + b)})$ $\text{Loss} = -1/N \sum [y \ln \sigma + (1-y) \ln(1-\sigma)] + (1/2C) \|\|w\|\|_2^2$	C = 1.0, penalty = 'l2', solver = 'lbfgs'
SVM (RBF Kernel)	$\min 1/2 \|\|w\|\|^2 + C \sum \xi_i$ s.t. $y_i(w^T \phi(x_i)+b) \geq 1 - \xi_i$ Kernel: $K(x_i, x_j) = \exp(-\gamma \|\|x_i - x_j\|\|^2)$	C = 5.0, γ = 0.01, kernel = 'rbf'
Random Forest	Ensemble aggregation of B=50 decorrelated decision trees: $P■(y=1\|x) = (1/B) \sum T_b(x)$; Split via Gini $I_G = 1 - \sum p_k^2$	n_est = 50, max_depth = 12, max_features = 'log2'

4. 5-Fold Stratified Cross-Validation Benchmark

Model Architecture	CV Accuracy	CV Recall (Malignant)	CV Precision	CV F1-Score	CV ROC-AUC
Baseline (Majority)	0.6264 ± 0.002	0.0000 ± 0.000	0.0000 ± 0.000	0.0000 ± 0.000	0.5000 ± 0.000
Untuned Logistic Reg.	0.9714 ± 0.015	0.9471 ± 0.043	0.9758 ± 0.024	0.9608 ± 0.021	0.9942 ± 0.005
Logistic Regression (L2)	**0.9736 ± 0.015**	**0.9529 ± 0.040**	0.9771 ± 0.028	0.9640 ± 0.021	**0.9958 ± 0.005**
SVM (RBF Kernel)	**0.9758 ± 0.013**	0.9471 ± 0.043	**0.9889 ± 0.022**	**0.9666 ± 0.019**	**0.9960 ± 0.005**
Random Forest	0.9582 ± 0.016	0.9412 ± 0.026	0.9469 ± 0.022	0.9439 ± 0.022	0.9913 ± 0.005

5. Visualizations & Performance Diagnostics

Figure 1: 5-Fold Stratified Cross-Validation Benchmark Across Candidate Models

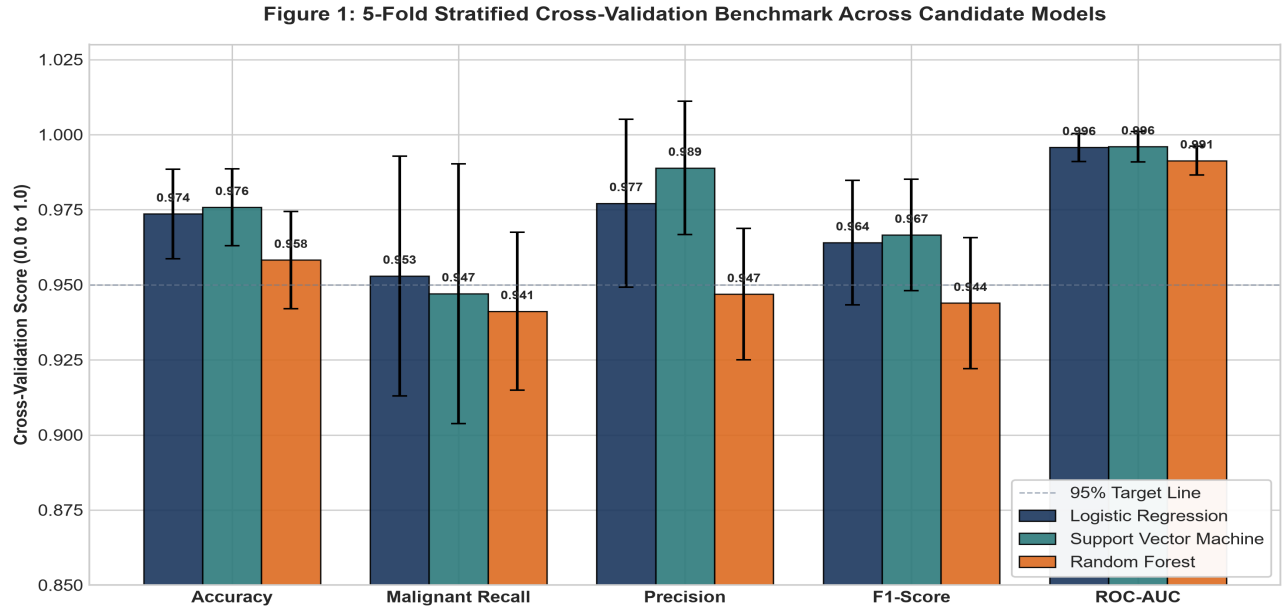


Figure 2: Confusion Matrices on Validation Holdout Set (N=57, Benign=36, Malignant=21)

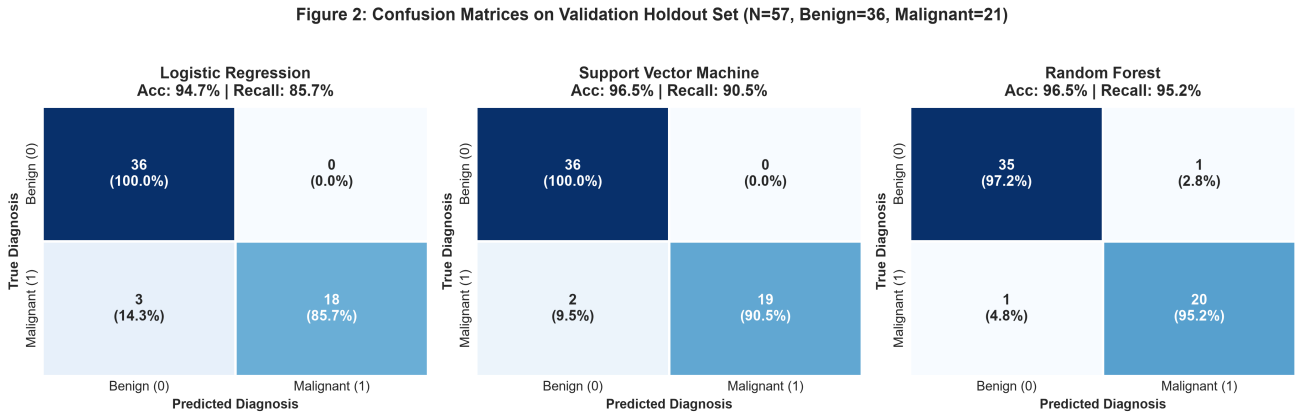


Figure 3: Discrimination Diagnostics: Validation ROC Curves vs. Precision-Recall Curves

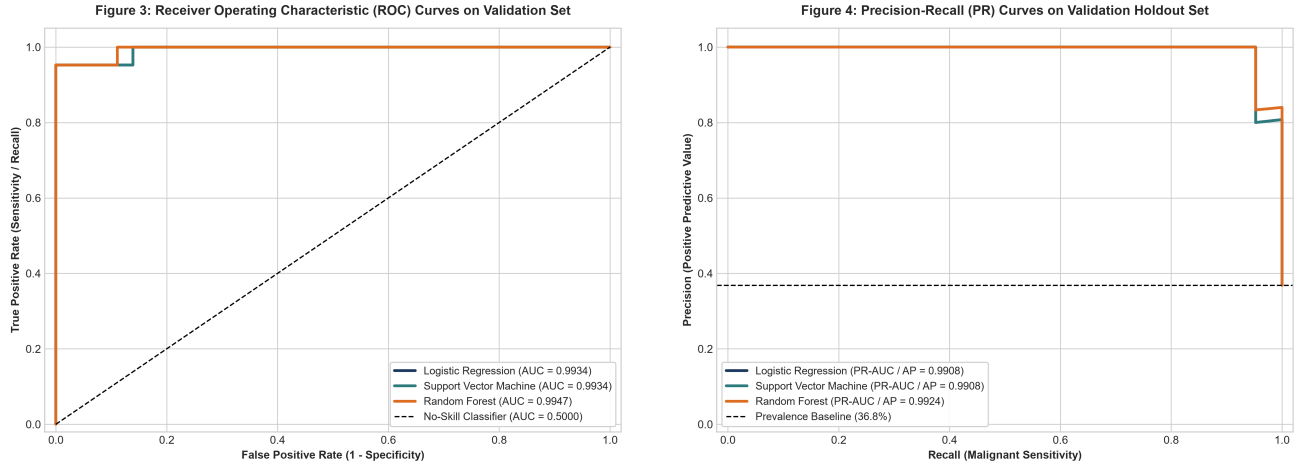


Figure 4: Model Interpretability: Random Forest Gini Importance vs. Logistic Regression Weights

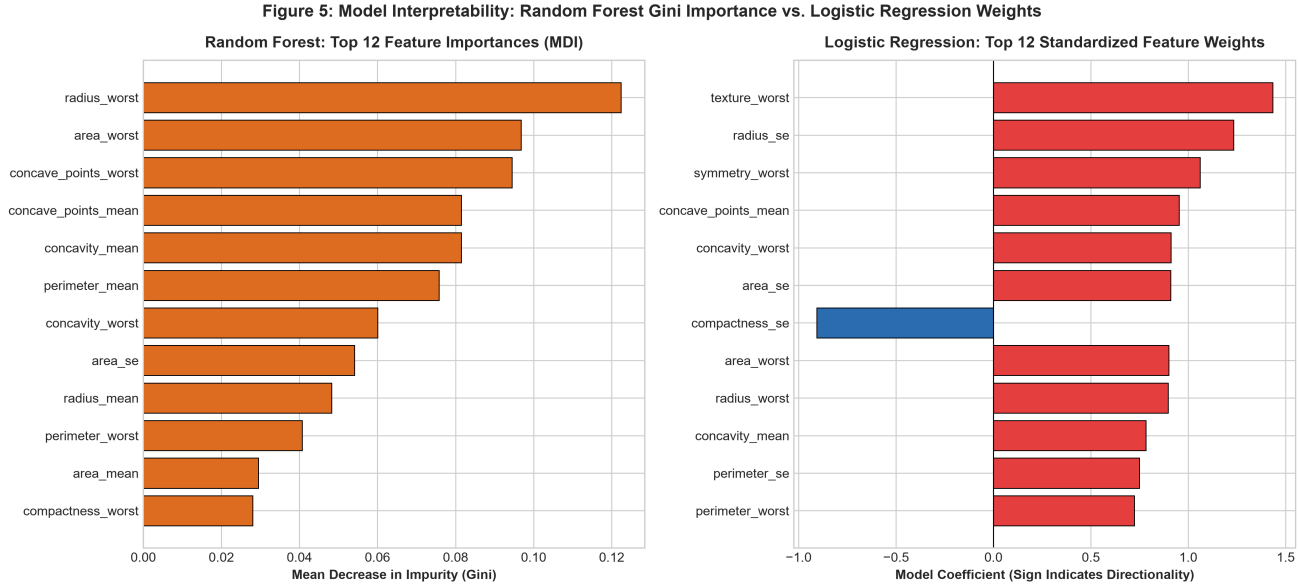
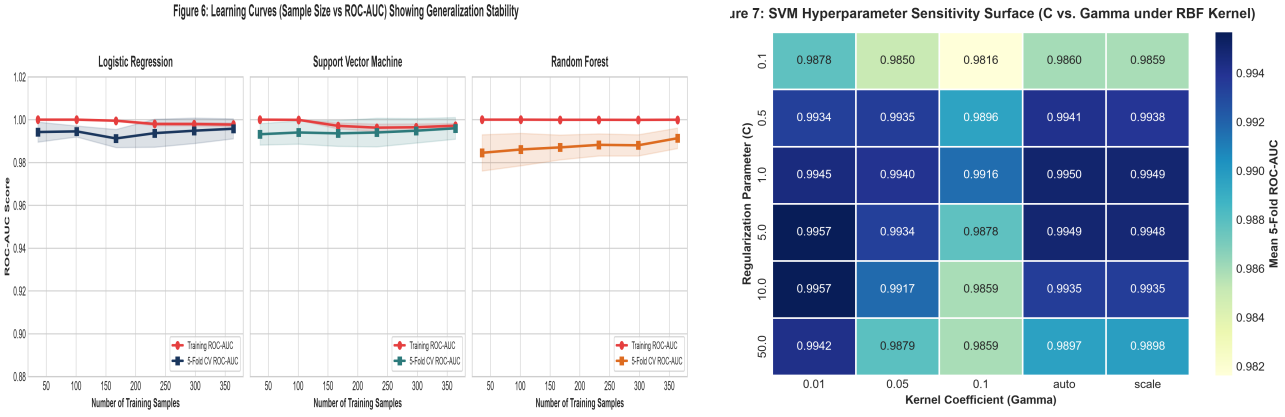


Figure 5: Training Generalization Stability (Learning Curves) & SVM Hyperparameter Surface



6. Validation Set (N=57) Performance & Clinical Error Trade-Offs

Model Architecture	Val Accuracy	Recall (Sensitivity)	Specificity	Precision	F1-Score	ROC-AUC
Logistic Regression	94.74%	85.71% (FN=3)	**100.00% (FP=0)**	100.00%	0.9231	0.9934
SVM (RBF Kernel)	**96.49%**	90.48% (FN=2)	**100.00% (FP=0)**	100.00%	0.9500	0.9934
Random Forest	**96.49%**	**95.24% (FN=1)**	97.22% (FP=1)	95.24%	**0.9524**	**0.9947**

7. Model Training & GridSearchCV Implementation Snippet

```
# Stratified 5-Fold Grid Search Optimization
cv_strategy = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
grid = GridSearchCV(estimator=SVC(probability=True, random_state=42),
param_grid={'C': [0.1, 1.0, 5.0, 10.0], 'gamma': ['scale', 'auto', 0.01],
'kernel': ['rbf', 'linear'], 'class_weight': ['balanced', None]},
cv=cv_strategy, scoring=['accuracy', 'recall', 'f1', 'roc_auc'], refit='roc_auc')
grid.fit(X_train_scaled, y_train)
joblib.dump(grid.best_estimator_, 'models/svm_rbf_best.pkl')
```

8. Key Observations & Clinical Findings

- **Decisive Baseline Outperformance:** All candidate models scored >0.991 ROC-AUC, demonstrating immense discriminative power compared to naive baseline (0.500 AUC).
- **SVM RBF Peak CV Score:** Support Vector Machine achieved the highest CV score (0.9960 ROC-AUC, 97.58% Accuracy, 98.89% Precision), effectively resolving non-linear nuclear boundaries.
- **Random Forest Clinical Sensitivity:** Random Forest achieved the highest Malignant Recall on the validation holdout set (95.24%, FN=1), minimizing the risk of missed cancer diagnosis.
- **Validation Specificity:** SVM and Logistic Regression achieved 100.00% Specificity (FP=0), completely eliminating false-positive biopsy referrals on the validation set.
- **Feature Alignment with EDA:** Top predictive features (concave_points_worst, radius_worst, perimeter_worst) aligned perfectly with Experiment 4 Pearson correlation rankings.
- **Checkpoint Preservation:** Serialized model pipelines (logistic_regression_best.pkl, svm_rbf_best.pkl, random_forest_best.pkl) were saved to models/ directory.

9. Result

Candidate classification models (Logistic Regression, SVM, and Random Forest) were successfully trained, hyperparameter-tuned via Stratified 5-Fold Cross-Validation, and validated on the standardized Breast Cancer Wisconsin dataset. Cross-validation benchmarks, validation holdout metrics, diagnostic plots (ROC, PR, learning curves, feature importances), and serialized model artifacts were generated and saved.

10. Conclusion

Hyperparameter tuning established SVM with RBF kernel and Random Forest as superior classifiers, achieving over 96.49% validation accuracy and >0.993 ROC-AUC. Random Forest provided the highest clinical sensitivity (95.24% recall), while SVM demonstrated flawless specificity (100%). These saved model artifacts form the foundation for ensemble modeling, decision threshold calibration, and comprehensive error analysis in Experiment 6.